

1 Part 18: Hypothesis Testing

2 Summary

3 **Statistical hypothesis testing** is a formal framework for “making discov-
4 eries.” It is ubiquitous in research, and overused. Nevertheless, it can be
5 useful, and it is definitely important to understand the concept.

6 In particular, one will see repeated references to **p-values** when reading
7 research. These are often misinterpreted.

8 Broadly speaking, hypothesis testing gives us a basis to make a claim that
9 an observed “feature” is “real” and not simply an artifact that occurred
10 due to random variability in our sample.

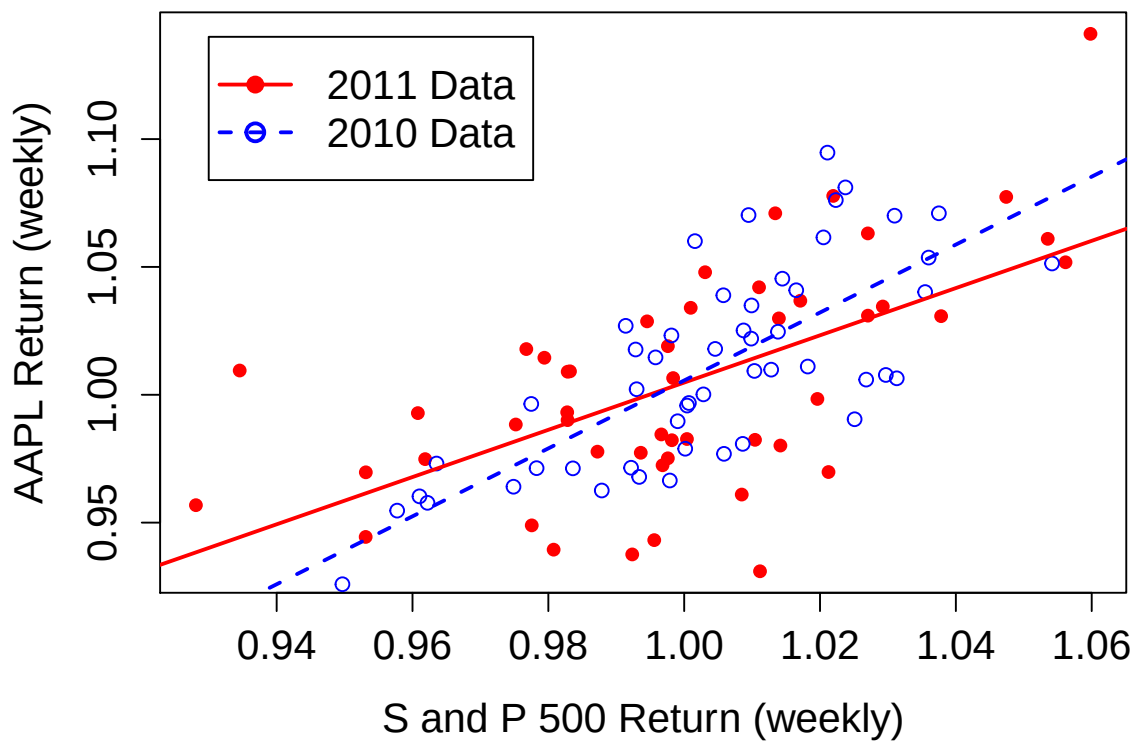
1 Consider the questions on the following three slides. In each case, the
2 question is not asking for the best estimate of a parameter, but instead
3 asking if the data provide sufficient evidence for some claim.

4 What do these examples have in common?

5 1. They each ask about a feature of the **population** from which the **sam-**
6 **ple** (observed data) is drawn.

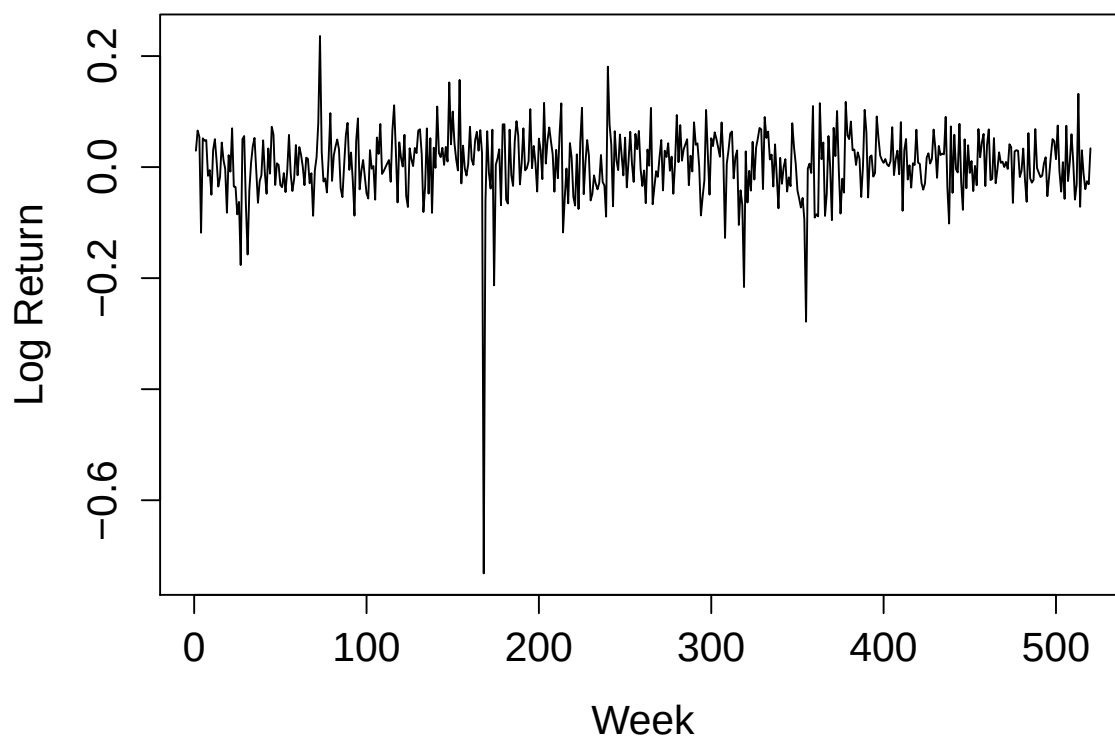
7 2. The answer to each question is definitely “No.” The issue is whether
8 or not the data provide **sufficient evidence** that the answer is “No.”

9 3. Each question puts forth a relatively simple “null hypothesis” and
10 asks “Is there sufficient evidence to reject the null?”



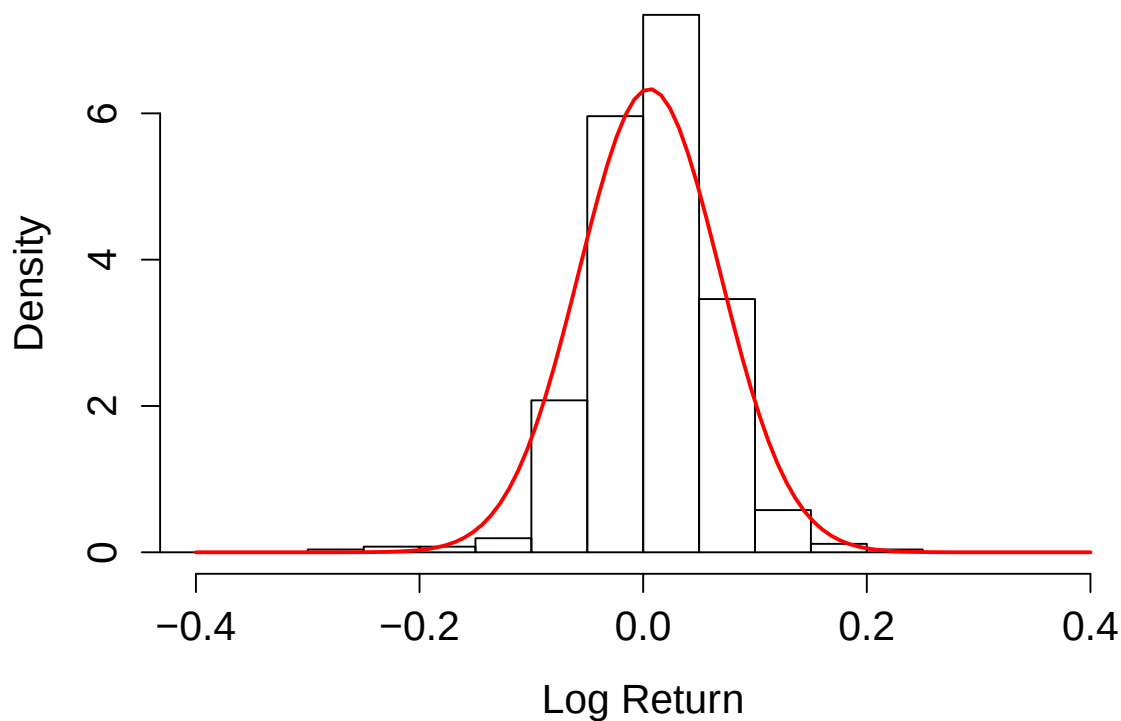
1

2 Is the slope of the regression line the same in 2010 and 2011?



1

2 Is the volatility constant over this time period?



1

2 Are these data normally distributed?

1 Such questions as these can be addressed using statistical hypothesis test-
2 ing.

3 The key requirement is that the question of interest, “Is there sufficient
4 evidence that...,” can be turned into a question regarding the population.

5 Often the hypothesis will take the form of a question regarding θ , but this
6 is not always the case.

1 Null and Alternative Hypotheses

2 The first two key ingredients to a hypothesis test are the **null hypothesis**
3 (H_0) and the **alternative hypothesis** (H_a or H_1).

4 **The alternative hypothesis is the claim you are testing to see if there is**
5 **strong enough evidence to support.**

6 The null hypothesis is the “baseline” or “status quo.”

7 The main thing to keep in mind is the following: **Statistical hypothesis**
8 **tests are designed to determine if there is strong evidence to support the**
9 **alternative (and reject the null). They are not designed to test if there is**
10 **strong evidence to support the null hypothesis.**

1 **Exercise:** A pharmaceutical company is testing a new treatment for a dis-
2 ease. The current treatment is 40% effective. Explain why a hypothesis
3 test is appropriate in this case. Formulate a reasonable null and alternative
4 hypothesis for the company.

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

11 _____

12

- 1 **Exercise:** Write out null and alternative hypotheses for each of the three
2 original examples. Be clear about what is being tested in each case.

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

11 _____

1 **The Test Statistic and the Rejection Region**

2 Not surprisingly, data are used to test the hypothesis. As before, random
3 variables $X_1, X_2, \dots, X_n$ are used to model the observable sample.

4 The **test statistic** is the function of the data that we will base our conclusion
5 on. Keep in mind that the test statistic is also a random variable.

6 Write $T = T(X_1, X_2, \dots, X_n)$.

7 The **rejection region** is the set of values for the test statistic such that the
8 conclusion is “reject the null hypothesis in favor of the alternative.” In
9 other words, when T falls into the rejection region, we conclude that there
10 is strong evidence in support of the alternative.

- 1 *How is the rejection region chosen?*
- 2 Consider the two possible errors we can make with a hypothesis test:
- 3 1. A **Type I error** is made when we reject the null even though the null is
- 4 true.
- 5 2. A **Type II error** is made when we fail to reject the null when the alter-
- 6 native is true.
- 7 The assumption is that we are **primarily concerned with making a Type I**
- 8 **error**. We want to limit the probability of making a Type I error.
- 9 We form the rejection region so that the probability of a Type I error is less
- 10 than or equal to some chosen amount α . This is often called the “size” or
- 11 “level” of the test.

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, with σ^2
- 2 known. We want to test $H_0 : \mu = 10$ versus $H_1 : \mu > 10$. A natural test
- 3 statistic in this case is $\bar{X}$. Formulate the rejection region for a level α test.
- 4 _____
- 5 _____
- 6 _____
- 7 _____
- 8 _____
- 9 _____
- 10 _____
- 11 _____
- 12 _____

1

2

3

4

5

6

7

8

9

10

11

12

1

2

3

4

5

6

7

8

9

10

11

12

1 In the previous exercise, we note that the result was based on knowledge
2 of the distribution of the test statistic under the null hypothesis. It is key
3 that the test statistic be chosen so that its distribution is known (at least
4 approximately), under the null. As usual, the central limit theorem can be
5 very helpful.

6 Also, we note that in the exercise we tested the **one-sided alternative hy-**
7 **pothesis** $H_1: \mu > 10$. In other cases we will test a **two-sided alternative**, for
8 example, $H_1: \mu \neq 10$. In such a case the null would be rejected for either
9 small or large values of the test statistic.

1 **Exercise:** Suppose that we tested $H_1: \mu \neq 10$ in the above example. What
2 would the rejection region be for fixed α ?

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

1 The p-value

2 The choice of α in the statistical hypothesis testing framework may seem
3 arbitrary. Why would one choose one value of α over another?

4 As α is chosen smaller, the experimenter is choosing to be more cautious,
5 to protect against Type I error. The opposite is true as α is chosen larger.

6 For a given data set, there is a **dividing line** between the choices of α for
7 which the conclusion is “reject the null” and the conclusion is “fail to reject
8 the null.” This dividing line is the **p-value**.

9 1. If $\alpha < \text{p-value}$, then “fail to reject the null.”

10 2. If $\alpha > \text{p-value}$, then “reject the null.”

1 It is important to note that the p-value is **a function of the data**.

2 The p-value is **not** the probability that the null hypothesis is true. Instead,
3 it is the largest choice of α such that the conclusion is “fail to reject the
4 null.”

5 The appeal is that one can report a p-value, and allow the reader to decide.
6 Most software packages report the results of hypothesis tests as p-values.

7 The next exercise illustrates another interpretation of the p-value: It is
8 the probability that, assuming the null hypothesis is true, the test statis-
9 tic would be equal to or more extreme in favor of the alternative than the
10 observed test statistic.

11 This interpretation is often useful for making computations.

- 1 **Exercise:** Return to our example when testing $H_0: \mu = 10$ versus $H_1: \mu >$
2 10 . Assume that $\sigma^2 = 1$. We observe $\bar{x} = 10.5$ with a sample size of $n = 50$.
3 What is the p-value in this case? What happens if we switch to $H_1: \mu \neq 10$?

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

11 _____

12 _____

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

11 _____

12 _____

1 The Wald Test

2 The **Wald Test** is built around a simple notion: An ideal choice for the test
3 statistic is the number of standard errors a parameter estimate is from the
4 null hypothesized value for that parameter.

5 Specifically, if testing $H_0: \theta = \theta_0$, then the test statistic is chosen as

$$T = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})}$$

7 T gives the “the number of standard errors $\hat{\theta}$ is from θ_0 .”

8 In a wide range of situations, T has approximately the standard normal
9 distribution under the null hypothesis. Hence, the region region is not
10 difficult to derive.

1 **Exercise:** What is one very important case in which we can state that T is
2 approximately standard normal under the null?

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

- 1 The three possible tests can be presented as follows:
- 2 1. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta > \theta_0$, then reject the null if $T > z_\alpha$.
 - 3 2. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta < \theta_0$, then reject the null if $T < z_\alpha$.
 - 4 3. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta \neq \theta_0$, then reject the null if either
 - 5 $T < z_{\alpha/2}$ or $T > z_{\alpha/2}$.

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli(p). Construct the
- 2 two-sided Wald Test for p . Suppose that $n = 100$ and $\sum X_i = 47$. Test
- 3 $H_0: p = 0.4$ versus $H_1: p \neq 0.4$. What is the p-value?

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

11 _____

12 _____

- 1 _____
- 2 _____
- 3 _____
- 4 _____
- 5 _____
- 6 _____
- 7 _____
- 8 _____
- 9 _____
- 10 _____
- 11 _____
- 12 _____

- 1 _____
- 2 _____
- 3 _____
- 4 _____
- 5 _____
- 6 _____
- 7 _____
- 8 _____
- 9 _____
- 10 _____

- 1 The Wald Test can be extended to testing multiple parameters simultane-
2 ously by noting the following result: If $\mathbf{X}$ is p -dimensional multivariate
3 normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, then

4
$$(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$$

- 5 has the chi-squared distribution with p degrees of freedom.

- 6 **Exercise:** Explain how this result can be used in conjunction with a multi-
7 parameter distribution when the MLE is used as the estimator.

8 _____

9 _____

10 _____

11 _____

12